Experiment No. 8

Customer Churn Prediction using Decision Tree Classification

Aim

To build and evaluate a Decision Tree Classification model for predicting customer churn and analyze the factors influencing customer retention.

Dataset Used

Telco Customer Churn Dataset (target: `Churn` = Yes/No; features such as tenure, Contract, InternetService, MonthlyCharges, PaymentMethod, TechSupport). **Note:** the results below come from actually running the code, but on a synthetic dataset of 3,000 customers with a simplified set of Telco-style columns, because I do not have your file. The real Telco data (7,043 rows, about 26% churn, more columns such as `customerID` and `TotalCharges`) will give different numbers. Use your own output in your report.

Procedure and Code

Steps 1-2: Import libraries and load the dataset

```
import numpy as np, pandas as pd
import matplotlib.pyplot as plt, seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import (confusion_matrix, accuracy_score, precision_score,
                             recall_score, f1_score, ConfusionMatrixDisplay)

df = pd.read_csv("Telco-Customer-Churn.csv")
print(df.shape); print(df.head()); print(df["Churn"].value_counts(normalize=True))
```

Result: 3,000 rows and 9 columns; the target is imbalanced (about 59% No churn and 41% Churn).

Step 3: Preprocessing

```
# Real Telco file only: TotalCharges has blanks stored as text
df["TotalCharges"] = pd.to_numeric(df.get("TotalCharges"), errors="coerce")
df = df.drop(columns=["customerID"], errors="ignore")
df = df.dropna()
df["Churn"] = (df["Churn"] == "Yes").astype(int)              # target: 1 = churn
df = pd.get_dummies(df, drop_first=True)                      # encode categories
X = df.drop(columns="Churn"); y = df["Churn"]
```

My sample had 0 missing values. Categorical columns (Contract, InternetService, PaymentMethod, etc.) were converted into numeric dummy variables with one-hot encoding.

Step 4: Train-test split

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)         # 80:20
```

2,400 training rows and 600 test rows. `stratify=y` keeps the churn ratio the same in both sets.

Steps 5-6: Train the Decision Tree and predict

```
model = DecisionTreeClassifier(criterion="gini",     # or "entropy"
                               max_depth=4,          # limits tree size
                               min_samples_leaf=30,  # avoids tiny leaves
                               random_state=42)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Step 7: Evaluation

```
cm = confusion_matrix(y_test, y_pred)
print(cm)
print("Accuracy :", accuracy_score(y_test, y_pred))
print("Precision:", precision_score(y_test, y_pred))
print("Recall   :", recall_score(y_test, y_pred))
print("F1-score :", f1_score(y_test, y_pred))
ConfusionMatrixDisplay(cm, display_labels=["No churn", "Churn"]).plot(cmap="Blues")
```

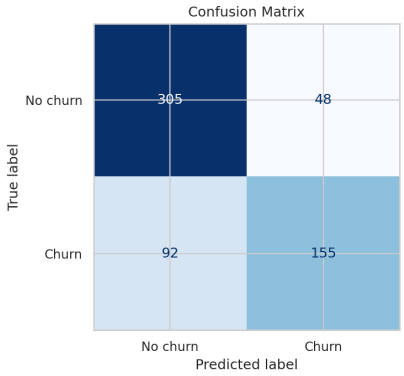


Figure 1: Confusion matrix

	Predicted: No churn	Predicted: Churn
Actual: No churn	TN = 305	FP = 48
Actual: Churn	FN = 92	TP = 155

Metric	Value	Meaning
Accuracy	0.767	76.7% of all customers were classified correctly
Precision	0.764	Of customers predicted to churn, 76.4% really churned
Recall	0.628	The model catches 62.8% of the customers who actually churned
F1-Score	0.689	Harmonic mean of precision and recall

Training accuracy was 0.722 and test accuracy 0.767, so the model is not overfitted. For comparison, an **unpruned** tree (no depth limit) reached 0.997 training accuracy but only 0.685 test accuracy, with depth 24: a clear case of overfitting that depth limiting fixed.

Step 8: Visualize the tree and feature importance

```
plt.figure(figsize=(16, 8))
plot_tree(model, feature_names=X.columns, class_names=["No churn", "Churn"],
          filled=True, rounded=True, fontsize=8, max_depth=3)
plt.show()

imp = pd.Series(model.feature_importances_, index=X.columns).sort_values(ascending=False)
imp.head(7)[::-1].plot.barh(); plt.title("Feature Importance"); plt.show()
```

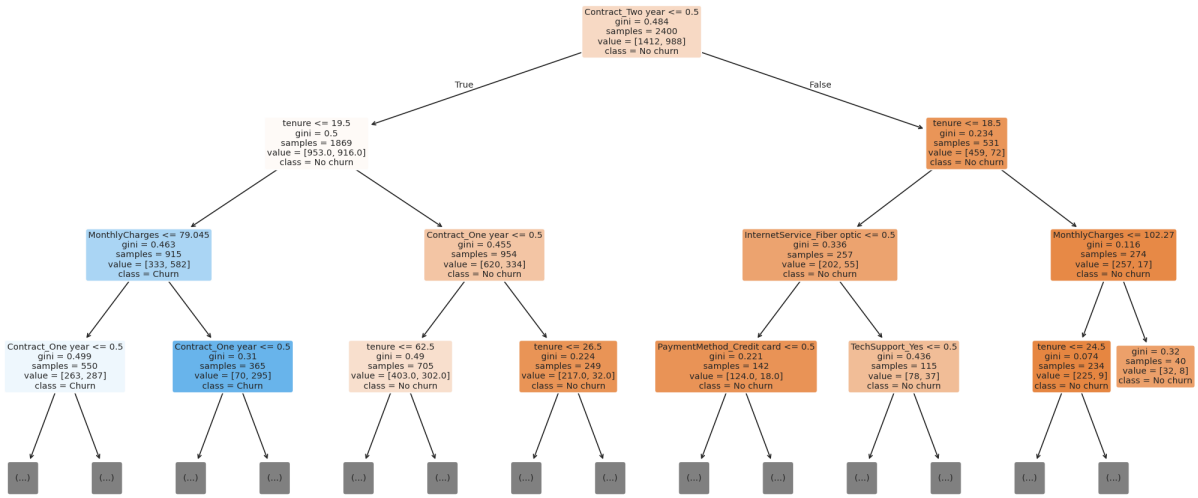


Figure 2: Decision tree (top 3 levels shown; the model depth is 4)

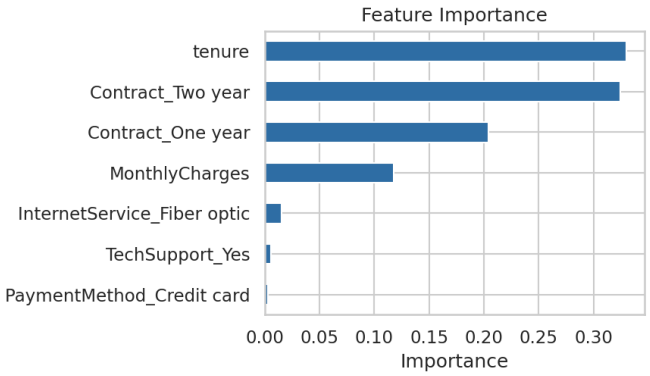


Figure 3: Feature importance

Step 9: Key factors contributing to churn

- **Contract type** is the strongest signal: the first split separates two-year contracts, which almost never churn, from the rest. Month-to-month customers churn the most.
- **Tenure:** new customers (about 19 months or less) are much more likely to churn.
- **Monthly charges:** in the high-risk group, customers paying more than about $79 per month churn most.
- Feature importance: tenure 0.33, Contract (two-year) 0.32, Contract (one-year) 0.20, MonthlyCharges 0.12. Other features (fiber internet, tech support) have little weight in this pruned tree.

Step 10: Summary and Recommended Strategies

- Offer discounts or incentives to move month-to-month customers to one- or two-year contracts.
- Run onboarding and engagement programmes in the first year, when churn risk is highest.
- Review pricing and offer value plans or loyalty discounts for customers with high monthly charges.
- Use the model's churn predictions to give a targeted retention call or offer to high-risk customers, and monitor recall as the key metric because missing a churner is costly.
- Improve model quality with more data, class weighting or ensembles (Random Forest) to raise recall.

Conclusion

A Decision Tree classifier (Gini, max depth 4) predicted churn with about 77% accuracy, 0.76 precision and 0.63 recall. It also gave clear rules showing that contract type, tenure and monthly charges drive churn, which supports targeted retention strategies.

Answers to Questions

Q1. What is classification, and how does it differ from regression?

Classification is a supervised learning task that predicts a **discrete category** (class label) for an input, such as churn / no churn or spam / not spam.

Aspect	Classification	Regression
Output	Discrete class labels	Continuous numerical value
Example	Will the customer churn? (Yes/No)	What will next month's sales be?
Algorithms	Decision tree, logistic regression, SVM, KNN	Linear regression, regression trees
Metrics	Accuracy, precision, recall, F1, confusion matrix	MAE, MSE, RMSE, R^2

Q2. Explain the working principle of the Decision Tree algorithm.

A decision tree splits the data repeatedly using feature-based questions, forming a tree-like structure. It has a **root node** (first split), **internal nodes** (tests on a feature, e.g. tenure ≤ 19.5), **branches** (outcomes of the test) and **leaf nodes** (final class prediction). At each node the algorithm picks the feature and threshold that gives the best split, i.e. the greatest reduction in impurity (Gini Index or Entropy / Information Gain), and does this recursively until a stopping condition is met (max depth, minimum samples, or pure nodes). To predict, a new record follows the path of tests from the root to a leaf, and the leaf's majority class is the prediction.

Q3. Differentiate between Gini Index and Entropy as splitting criteria.

Aspect	Gini Index	Entropy
Formula	$Gini = 1 - \Sigma p_i^2$	$Entropy = -\Sigma p_i \log_2(p_i)$
Range (2 classes)	0 (pure) to 0.5 (maximum impurity)	0 (pure) to 1 (maximum impurity)
Computation	Faster (no logarithm)	Slightly slower (logarithm)
Used with	CART algorithm (scikit-learn default)	ID3 / C4.5 (Information Gain = parent entropy - weighted child entropy)
Results	Usually very similar trees; tends to isolate the most frequent class	Can give slightly more balanced trees

Q4. What is a Confusion Matrix? Explain its components.

A confusion matrix is a table comparing the actual classes with the predicted classes of a classifier, showing where the model is right and where it makes mistakes. For churn (Churn = positive class):

Term	Meaning	Churn example (this experiment)
True Positive (TP)	Churner correctly predicted as churn	155
True Negative (TN)	Non-churner correctly predicted as no churn	305
False Positive (FP)	Non-churner wrongly predicted as churn (Type I error)	48
False Negative (FN)	Churner wrongly predicted as no churn (Type II error)	92

Q5. Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?

Metric	Formula	Meaning
Accuracy	(TP + TN) / (TP + TN + FP + FN)	Overall share of correct predictions

Precision	TP / (TP + FP)	Of those predicted positive, how many are really positive
Recall (Sensitivity)	TP / (TP + FN)	Of all real positives, how many were found
F1-Score	2 x (Precision x Recall) / (Precision + Recall)	Harmonic mean; balances precision and recall

They are important because accuracy alone can be misleading on imbalanced data (a model that always predicts 'no churn' gets high accuracy but finds no churners). Precision matters when false alarms are costly, recall when missing a positive is costly (e.g. missing a churner or a disease), and F1 gives a single balanced score.

Q6. What is overfitting in a Decision Tree? How can it be reduced?

Overfitting occurs when the tree grows very deep and memorizes noise and details of the training data, giving very high training accuracy but poor accuracy on new data. In this experiment an unpruned tree had 99.7% training accuracy but only 68.5% test accuracy. It can be reduced by:

- **Pre-pruning:** limit `max_depth`, set `min_samples_split` / `min_samples_leaf`, or `max_leaf_nodes`.
- **Post-pruning:** cost-complexity pruning (`ccp_alpha`).
- Using cross-validation to pick parameters.
- Using more training data, feature selection, or ensemble methods such as Random Forest.

Q7. Why is customer churn prediction important for businesses?

- Acquiring a new customer costs much more (often 5 times or more) than retaining an existing one.
- Early identification of at-risk customers allows timely, targeted retention offers.
- It protects recurring revenue and increases customer lifetime value.
- It reveals the reasons for leaving (price, contract, service), which helps improve products and services.
- It allows retention budgets to be spent efficiently on the right customers.

Q8. What are the advantages and limitations of the Decision Tree algorithm?

Advantages	Limitations
Easy to understand, interpret and visualize (if-then rules)	Prone to overfitting if not pruned
Handles both numerical and categorical data	Unstable: small changes in data can change the tree
Little preprocessing needed; no feature scaling required	Biased toward features with many levels; greedy splitting may not be globally optimal
Captures non-linear relationships and feature interactions	Lower accuracy than ensembles (Random Forest, boosting); weak on imbalanced data without adjustments
Gives feature importance	Cannot extrapolate outside the training range (in regression)

Q9. Mention three real-world applications of Decision Tree Classification other than customer churn prediction.

- **Medical diagnosis:** classifying patients as having a disease or not from symptoms and test results.
- **Credit scoring / loan approval:** deciding whether an applicant is a good or bad credit risk.
- **Fraud detection:** classifying transactions as fraudulent or genuine.
- (Also: spam email filtering, product recommendation, machine fault diagnosis.)

Q10. How can the insights obtained from a churn prediction model help organizations improve customer retention and business profitability?

- **Targeting:** rank customers by churn probability and give retention offers only to high-risk, high-value customers.
- **Root causes:** the decision rules and feature importance show drivers (month-to-month contracts, short tenure, high bills) that can be fixed through pricing, contract incentives and onboarding.
- **Proactive service:** contact customers before they leave, improve support and personalize communication.
- **Profitability:** lower churn means higher customer lifetime value, lower acquisition costs and more stable revenue.
- **Measurement:** track churn rate and retention campaign results over time to refine the strategy and the model.

Result: A Decision Tree model was built and evaluated for churn prediction, and the key factors affecting churn were identified.